

Carnivore2 User Guide

English version

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Русская версия находится здесь: [Инструкция по эксплуатации Carnivore2](#)

About the project

Carnivore2 is the multi-functional cartridge for the MSX platform that was created in 2017 by RBSC. The project is still supported — new firmware for FPGA is being released, as well as the software — the tools for working with the cartridge, the Boot Menu (boot block), etc. The website of RBSC can be found [here](#).





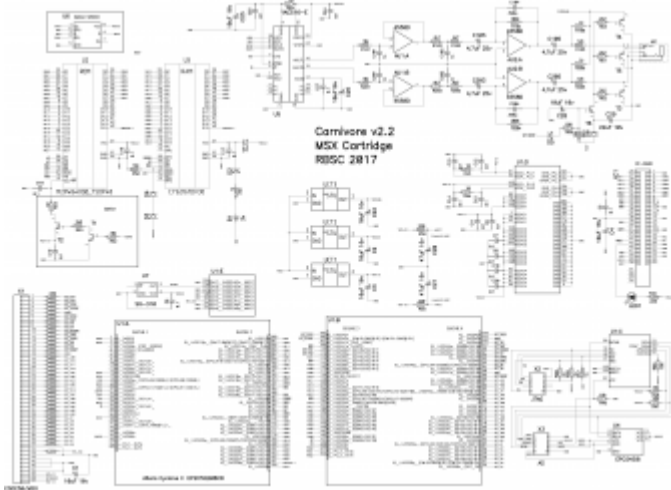
Carnivore2 features

- External storage: CF card (CompactFlash)
 - Nextor is used as DOS (built-in support for FAT12/16, maximum partition size: 4 GB)
 - High read and write speeds
 - Supports SD and MicroSD card adapters
 - Nextor supports floppy disk emulation with DSK files
 - Utilities compatible with MSX-DOS versions 1 and 2
 - The cartridge can be configured as a RAM extension, IDE disk, FMPAC and SCC/SCC+ sound cards, or a combination of these devices
- RAM: 2048 KB (2 MB)
 - Includes:
 - 1024 KB main RAM with mapper
 - 256 KB for ROM shadowing

- 720 KB additional RAM with a mapper, similar to MegaRAM
 - 4 KB for the FMPAC SRAM (a backup battery is needed to save data after turning off the power)
- Flash memory (FlashROM): 8 MB capacity, 64 Mbit/s
 - The first 256 KB are used for service information and ROM BIOSes
 - Mapper emulation:
 - Linear 64 KB mode
 - ASCII8
 - ASCII16
 - Konami4
 - Konami5 (SCC/SCC+)
 - Custom mapper
- Sound
 - PPI, PSG or Dual-PSG (I/O ports: 10h-11h or A0h-A1h)
 - Konami SCC and SCC+ emulation
 - OPLL emulation (YM2413, MSX - Music), BIOS IU translated to English
 - [Volume setting for all emulated audio devices](#)
 - PSG and PPI can be enabled and disabled in the user interface
- Additional features
 - Boot menu with a choice of games and configurations saved in flash memory
 - Sorting of directory records
 - [Customizable user interface](#) (colors, sorting, key repeat speed, fade effects)
 - [Help system](#)
 - Joystick and joypads support
 - [50/60Hz instant switching support](#)
 - [Selectable Turbo or R800 modes](#)
 - Auto-Start of configuration entries and ROMs with a startup delay
 - Allows to run 2 ROMs at the same time in the Dual-Slot screen
 - Mono/stereo modes for FMPAC output
 - Allows to set the default startup frequency (50 or 60Hz)
 - The device is designed in the form factor of the standard MSX cartridge
 - It operates at the standard frequency of 3.58 MHz, as well as turbo frequencies up to 7.11 MHz
 - Implemented on FPGA (EP2C5Q208C8 Altera Cyclone II)
 - Special software is used to control all functions
 - Download programs in ROM format to flash memory and RAM
 - [Backing up flash memory, configuration RAM settings, and FMPAC RAM contents](#)
 - [Program for testing the IDE interface](#), etc
 - Easy setup and user-friendly interface

Links to important Carnivore2 documents and websites

The below table contains the links to important Carnivore2 documentation and websites.

 CF card and adapter compatibility reference	List of compatible cards and adapters
 Detailed technical description of the cartridge	Internal blocks and data layout
 Changelog	Version-based list of changes
 Partslist	List of components and their nominals
 Circuit diagram (clickable) 	Schematics
 Link to official site	RBSC's website

 Repository on GitHub	Source code, board files, documentation
 Article on MSX Wiki	Official MSX Wiki article

List of authorized distributors

The below table contains the list of authorized Carnivore2 manufacturers and distributors.

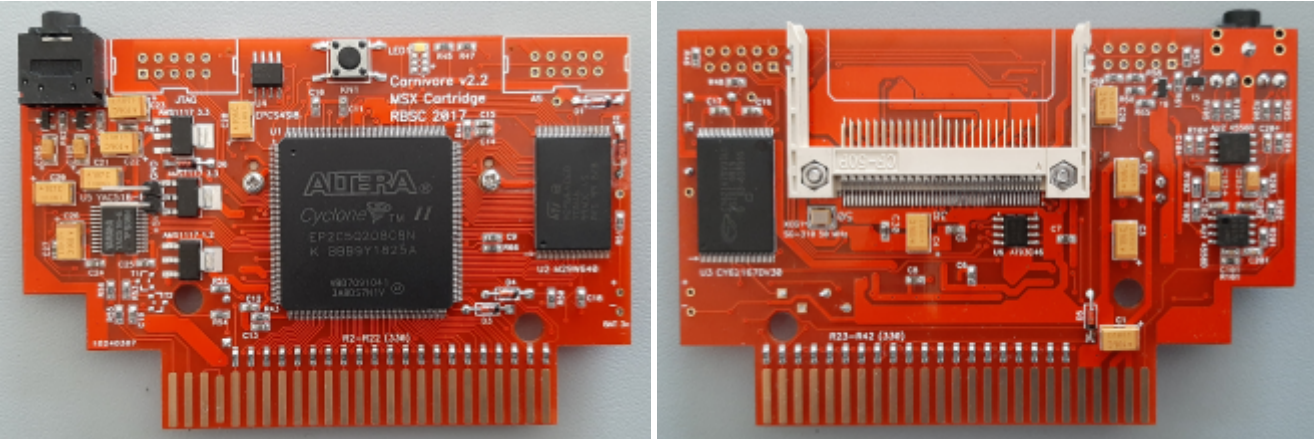
Order from Maxioli (RBSC) webshop Order from Maxioli's forum Contact the seller	Official distributor in Russia
Purchase from HobbyRetro online store	Official distributor in Spain
Purchase from 8bits4ever online store	Official distributor in Spain
Purchase from Retro Game Restore online store	Official distributor in Taiwan
Purchase from Carmeloco	Official distributor in Spain
Purchase from The Retro Hacker Store	Official distributor in Brazil
Purchase from Denjhang's Retro Hardware	Official distributor in China
Purchase from Retro Tech Store	Official distributor in Brazil

Photos of the cartridge board

The board from a test batch, assembled by RBSC:



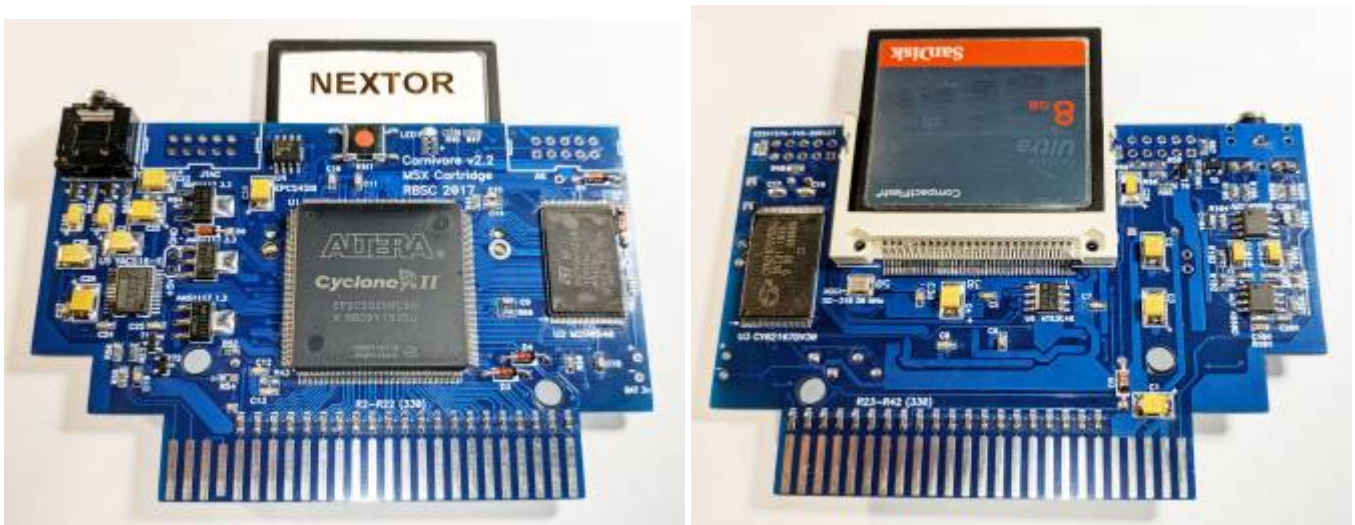
The board assembled by 8bits4ever (authorized manufacturer/seller, Spain):



The board assembled by Maxioli (authorized manufacturer/seller, Russia):



The board assembled by Carmeloco (authorized manufacturer/seller, Spain):



The board assembled by Retro Game Restore (authorized manufacturer/seller, Taiwan):



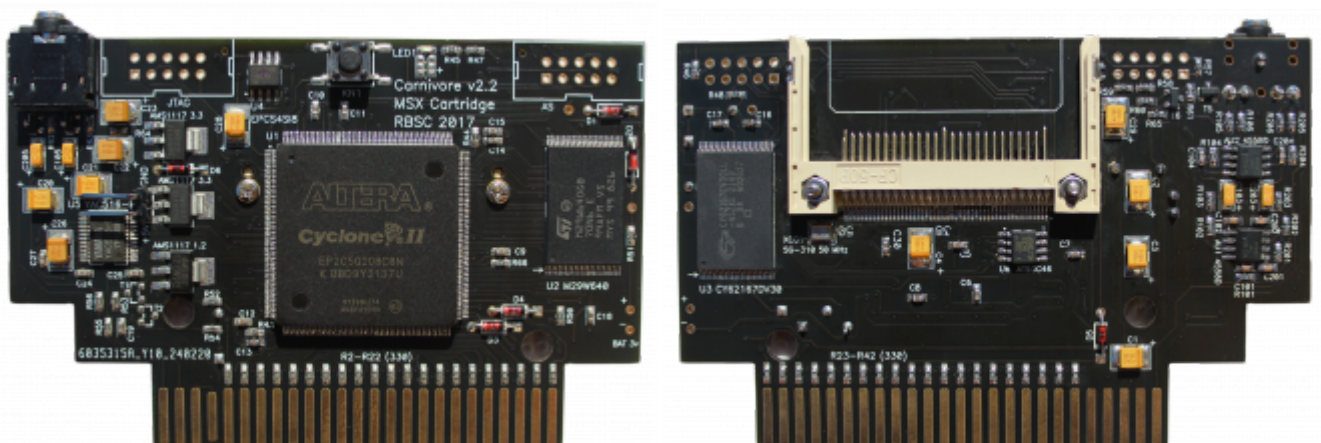
The board assembled by The Retro Hacker Store (authorized manufacturer/seller, Brazil):



The board assembled by the Denjhang's Retro Hardware Store (authorized manufacturer/seller, China):



The board assembled by the Retro Tech Store:



Note: on the board produced by Maxioli some tantalum capacitors were replaced with ceramic ones, also the audio socket was moved to the back of the board.

Note: The boards of the first batch made by the Retro Tech Store were not gold-plated. All next batches have gold-plated boards.

Important information





WARNING! To avoid damage to the Carnivore2 cartridge and your MSX computer hardware never insert or remove the cartridge when a computer is powered on! Always power off your computer before inserting or removing any cartridges!



If any of your CF cards or SD-to-CF adapters no longer work with Carnivore2, try to select the card or the adapter using the below linked compatibility document.



CF card and adapter compatibility reference



The correct operation of the Carnivore2 cartridge is not guaranteed in the R800 mode on Panasonic Turbo-R computers. There may be various anomalies in this mode, for example the games that require a system restart as well as the configuration entries won't work correctly. So for such games and configuration changes it's not recommended to enable the R800 mode in the Boot Menu. Certain features of Carnivore2 may not work correctly on computers with less than 16kb of RAM (for example on Casio PV-7).

In some rare cases Carnivore2 may not work correctly (according to our info). Such anomalous behavior has been spotted on at least 2 rare computers: Mistubishi ML-G30 and Toshiba HX-34. If you have such computers and you are able to boot them with Carnivore2, please inform the RBSC's coordinator.



Carnivore2 is incompatible with Yamaha's network modules:

- network module [KYBT](#) (installed into [Yamaha YIS-503IIR](#))
- network module [KYBT2](#) — [Yamaha Serial I/O Unit \(CB55448\)](#) (installed in [Yamaha YIS-503IIR](#) and [Yamaha YIS-805-128R2](#))

If those modules are detected, Carnivore will show a warning message and halt a system to prevent conflicts. In this case you need to remove the network module from a computer and also remove the network ROM cartridge from the front slot of the YIS-503II MSX1.



Certain MSX computers activate the built-in software after power-on and this prevents Carnivore2's Boot Menu from being started. To bypass the built-in software it's usually enough to hold the defined key during a computer's startup. For example Panasonic A1 (as well as on A1 MK2), you need to hold the [DEL](#) key until the Boot Menu appears or MSX-DOS is loaded. In May 2023 the special [BYPASSFW.ROM](#) file has been added into the release in order to simplify bypassing of the built-in firmware.



Certain games with the so-called “delayed start”, for example Metal Gear 2 and King's Valley 2, will not work on the [Yamaha YIS-503IIR](#) because of incompatibility with the built-in [CP/M](#). It is advised to removed the CP/M from the SubROM on these computers, by writing [this file](#) into the 27c256 chip and replacing the Subrom chip on the board. Alternatively you can use a different file with Subrom and TESTRAM. This file can be downloaded from [here](#).



On several MSX computers (for example on Casio PV models) as well as on several MSX clones (for example on Omega) there's no +12v and -12v power on the cartridge slots. This power is needed for the audio amplifier that is installed on Carnivore2 cartridge's board. Without this power Carnivore2 will not be able to play FMPAC,



SCC and PSG music and sound effects.



When a computer is just powered on with the Carnivore2 cartridge inserted into a slot, it may reboot twice. This is normal and was implemented to make sure that the cartridge is fully initialized after the cold boot. You can enable the dual-reboot feature in the Configuration settings.



The [Boot Menu](#) can detect that it is running on Korean or Arabic MSXs and in case it is not compatible with those systems, it will output a message and will try to boot to DOS after 10 seconds. In such a case a compatible Boot Menu (BOOTCMFC.BIN) should be installed into the cartridge (see the "Special" subfolder in the repository for the compatible version of the boot menu).

The Setup



If you purchased the already assembled and configured cartridge, you don't need to do the firmware and BIOS uploading.

After assembling, the cartridge needs to be programmed in order to function properly. The following steps are necessary:

1. [Upload the Altera's firmware](#)
2. Initialize the directory
3. Write the Boot Menu
4. Write the IDE BIOS
5. Write the FMPAC BIOS
6. Restart MSX

 The below described features are supported in the Boot Menu and FPGA firmware starting from version 2.40; in the older versions, these features may be missing or have different functionality.

How to prepare the CF card

The CF card needs to be formatted in the [FDISK](#) program and needs Nextor's files to be copied onto it.



Partitioning of CF memory card must be done in Carnivore2 cartridge!

1. Insert CF memory card into the cartridge and start the computer
2. When the boot menu appears, press the [Esc](#) button to boot using the default configuration
3. The computer boots into the MSX-BASIC
4. Type CALL FDISK and press [Enter](#) to run the partitioning program
5. Create one or more partitions using the [FDISK](#) interface
6. When finished, restart the computer
7. You need to write the Nextor system files so that you can boot into MSX-DOS2 from the card. To do this, copy the files from the archive or disk image that are available [here](#) and [here \(as disk image\)](#) into the root directory of the CF card. Then copy the NEXTOR.SYS file from [this location](#) into the root directory of the CF card

8. Example of copying files from a floppy disk to the main section of a CF memory card using a GoTek-based drive emulator:

```
copy "b:\*.*" to "a:\"
```

- a: — mounted CF card's partition
- b: — another drive (GoTek emulator, a physical 720kb drive, a disk of another IDE controller)

How to upload the firmware



Before uploading the firmware please make sure that the CF card is not inserted!

For updating/uploading of the firmware you will need:

1. [Quartus II Web Edition \(Free\) 15.0](#) software
2. Byte Blaster or USB Blaster programmer (can be purchased on Ebay or AliExpress)

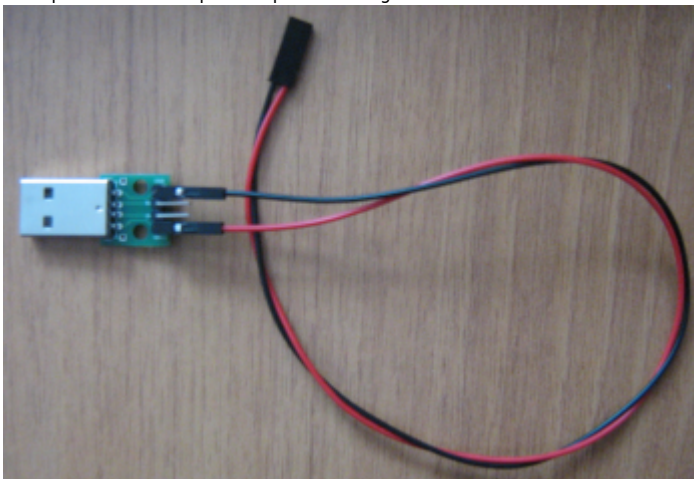
From the Individual Files tab please [download](#) and install the following software:

- Quartus II Software (includes Nios II EDS)
- Cyclone IV device support

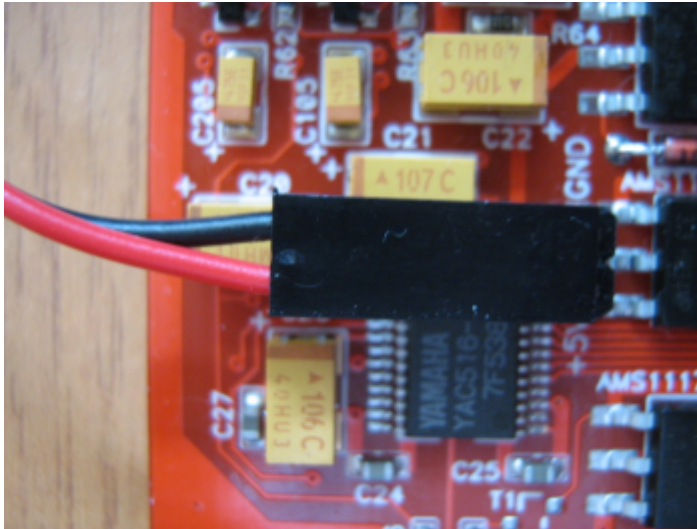
The following actions need to be performed:

1. Connect the USB Blaster programmer to the PC
2. Run the program Quartus II Web Edition
3. Select menu item Tools/Programmer
4. In the Programmer window:
 - Click **Hardware setup**
 - in the Hardware setup → Hardware Settings → Currently selected hardware choose your programmer device
 - Click **Close**
 - The device's name will be displayed to the right of the **Hardware setup** button and to the right of the Mode field, there you need to select Active Serial Programming
 - Click **Add Device**
 - in the Select Devices chose the EPCS4 device
 - Click **Ok**
 - Right click on the device, chose Change File and select the desired firmware file with the .pof extension from your hard disk
 - Make sure to check these options:
 - ✓ Program/Configure
 - ✓ Verify
 - ✓ Blank-Check
 - Supply 5 volts to the cartridge board (mind the polarity!)

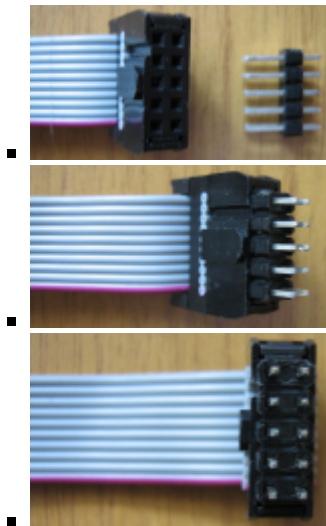
A simple USB cable to power-up the cartridge:



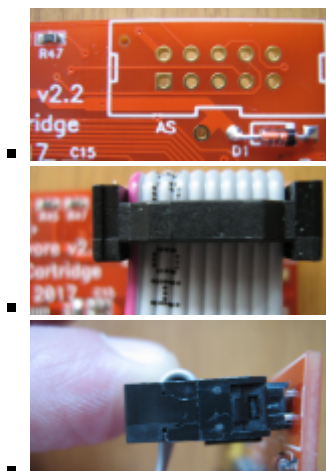
Power cable connected to the cartridge:



- Connect the programmer's cable to the cartridge's AS connector (make sure the cable is connected correctly!)
To connect the programmer to the AS connector on the cartridge, you need a dual-row pin header (PLD 2 × 5, 2.54 mm) that must be inserted into the programmer cable's connector:

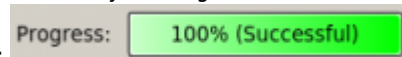


Then the programmer's connector must be inserted into the AS connector of the cartridge and slightly tilted to ensure good contact between the pins and the board:



5. Click Start and follow the programming and verification process while firmly holding the connector. After successful

completion of the firmware uploading you should get this message:



6. When the operation is complete, disconnect the cables from the cartridge in the following order:
 - programmer cable from the AS connector
 - power cable



Command line based Carnivore2 firmware upgrading is described [here](#)

How to enable the cartridge and install BIOS ROMs



Please always use the latest version of the C2MAN and C2MAN40 utilities to write necessary software into the cartridge!



When loading the Boot Menu and BIOSes into the cartridge on MSX Turbo R, please disable the R800 mode!

The freshly-assembled Carnivore2 cartridge will not boot to MSX-DOS2 without the specific software that needs to be loaded into the FlashROM chip from any disk drive.

Insert the cartridge into the MSX slot, preferably into the first slot. Power up MSX and check if it works well. If the computer doesn't work properly, remove and inspect the cartridge. To fully configure the cartridge the following actions need to be performed:

1. Make sure that all 3 BIN files (BIDECMFC.BIN, B00TCMFC.BIN, FMPCCMFC.BIN) are in the same folder with the utilities
2. Run the [C2MAN.COM](#) or [C2MAN40.COM](#) (for MSX1 only) utility
3. When asked, enter the slot number where the cartridge is located (for example "10" for first slot, "20" for second slot, etc.)
4. From the [main menu](#) select "Open cartridge's Service Menu" using the [9](#) key
5. With the [7](#) key select "[Fully erase FlashROM chip](#)" and confirm twice; if you are only updating the cartridge, this step may be skipped
6. With the [3](#) key select "[Init/Erase all directory entries](#)" to initialize the directory
7. With the [4](#) key select "[Write Boot Menu \(BOOTCMFC.BIN\)](#)" to write the Boot Menu
8. With the [5](#) key select "[Write IDE ROM BIOS \(BIDECMFC.BIN\)](#)" to write Nextor IDE BIOS
9. With the [6](#) key select "[Write FMPAC ROM BIOS \(FMPCCMFC.BIN\)](#)" to write the English FMPAC BIOS
10. If there were no errors during the steps 5-9, then power down and start your MSX

Writing BIOS and Boot Menu on a diskless computer

There's also another way to write the Boot Menu and BIOSes into the FlashROM chip. This can be done on a diskless MSX computer by loading the necessary files via the cassette interface and copying them from the specially formatted CF card into the FlashROM chip.

For more information please see this [file](#). After CF2FLASH utility you can use the Carnivore2.rom file from the OpenMSX folder to write the latest available Boot Menu and BIOS versions into the FlashROM chip. So, there's no longer need to constantly update the CFimage.bin file.

When writing contents of Carnivore2.rom or CFimage.bin onto a CF card, you need to be extremely careful not to overwrite the contents of your local hard drives (destroying data on them). It's recommended to use a separate CF card for this operation as its contents will be destroyed.

Usage instructions

The program that serves as the main interface of the Carnivore2 cartridge after a computer is powered on is called the [Boot Menu](#). It can be also referenced as "Boot Block" or "bootblock".



If the cartridge is inserted into slot 3, the system will be halted immediately. In this case please power down the system and insert the cartridge into another free slot..

If the Boot Menu version is incompatible with the MSX computer, the following warning message will be shown and the cartridge will try to boot into MSX-DOS2 within the next 10 seconds. In this case please write the special version of the Boot Menu into the cartridge (see the Special folder in the repository).

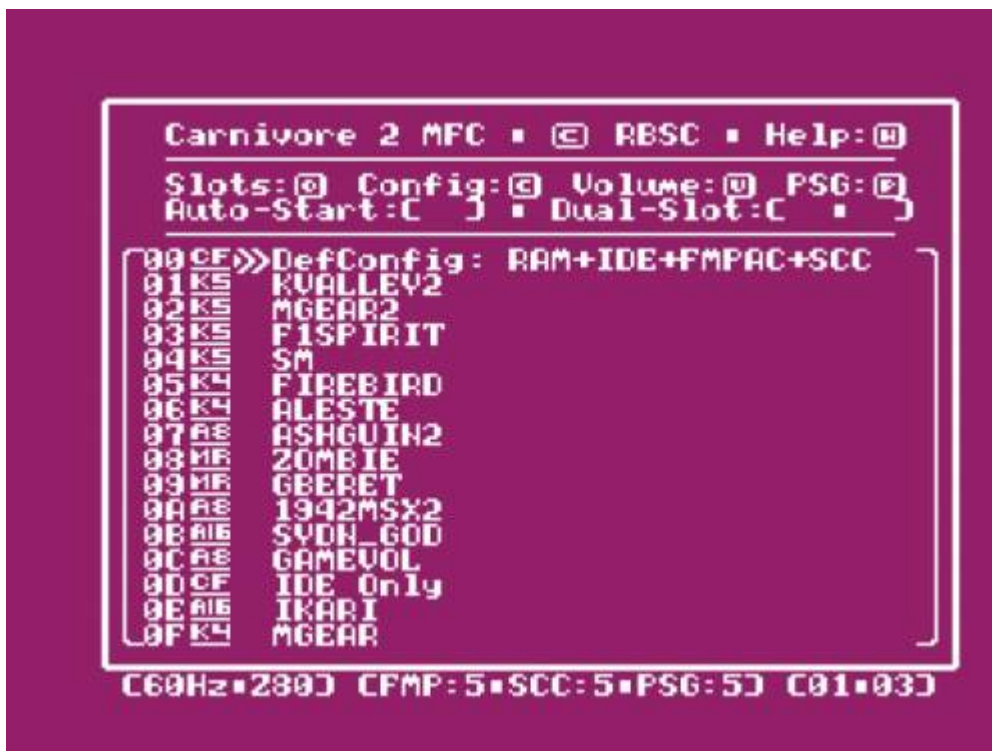


If there's a network module from KYBT or KYBT2 system (Yamaha-based computer classes that were distributed in USSR), the following warning message will be shown and the system will be halted. In this case please remove the network module from the computer.

```
KVBT/KVBT2 network module found!
Remove network module to prevent
conflicts with Carnivore2 cartridge.
System halted!
```

How to work with Boot Menu

The Boot Menu allows to start the ROMs from the FlashROM chip and to restart the cartridge with the desired configuration. After a computer shows its boot logo, the cartridge's Boot Menu should appear and you should see the [main menu](#) with the list of directory entries.



Navigating the menu is very easy. Here are the key assignments:

	boot MSX using the default configuration: all enabled
	previous/next page

 	select ROM/CFG entry
<u>Space</u>	start selected entry (single slot configuration)
<u>G</u>	start an entry directly
<u>R</u>	reset MSX and start an entry
<u>Enter</u>  <u>O</u>	Dual-Slot setup page
<u>1</u>	select entry for the master slot
<u>2</u>	select entry for the slave slot
<u>A</u>	select entry for autostart
<u>D</u>	clear Auto-Start & Dual-Slot settings
<u>F</u>	toggle 50Hz or 60Hz frequency
<u>T</u>	toggle Turbo or R800 mode
<u>C</u>	customize configuration
<u>P</u>	setup PSG and PPI Clicker
<u>V</u>	volume settings of FMPAC and SCC sound cards
<u>H</u>	help
<u>L</u>	jump to the last used directory entry
<u>M</u>	toggle background music playback
<u>S</u>	toggle help scroller

The main menu also supports the built-in joypads and external joysticks connected to any of the 2 joystick ports. The joystick's stick movements and pressing the buttons are interpreted as follows:

 	same as cursor keys
 	same as cursor keys
<u>A</u>	start an entry (same as <u>Space</u> on keyboard)
<u>B</u>	exit from Boot Menu (same as <u>Esc</u> on keyboard)

All other joystick directions are ignored.

The following data is displayed on the status line of the screen:

- [VDP Frequency](#)
- [CPU mode](#)
- [volume](#):
 - FMPAC
 - SCC
 - PSG
- Directory page numbers, in the following format: CP•TP
 - CP — current page
 - TP — total pages

Directory entry icons

The symbols that are displayed near the names of directory entries indicate the following:

K5	Konami5 (SCC) mapper
K4	Konami4 mapper
A8	ASCII8 mapper
A16	ASCII16 mapper
MR	mini ROM (8, 16, 32, 48 and 64kb ROM without mapper)
CF	configuration entry

UN	unknown mapper
–	for other mappers

On Arabic and Korean MSXs there will be the following indication (prior to Boot Menu v2.40):

K	Konami5 (SCC) mapper
k	Konami4 mapper
a	ASCII8 mapper
A	ASCII16 mapper
M	mini ROM (8, 16, 32, 48 and 64kb ROM without mapper)
C	configuration entry
U	unknown mapper

Please keep in mind that some ROMs may require alternative starting method, so if pressing Space doesn't start the ROM, try starting the ROM directly or after reboot.

Auto-Start

The Auto-Start option allows to start any configuration entry or ROM image when a computer is powered on. Starting from version 2.40 of the Boot Menu, the record number for the auto-start feature is saved into the configuration EEPROM, so this chip must be installed on the Carnivore2's board. The selection of an entry for auto-start is done with the A key from the main menu. To cancel auto-start for an entry, press the D key.

There are several keys that can affect the cartridge's functionality at boot level. Certain keys can cancel the auto-start, other keys can skip the Boot Menu:

<u>F3</u>	use default UI settings
<u>F4</u>	cancel auto-start
<u>F5</u>	skip Boot Menu



When auto-start is set for any entry, after reboot a message will be shown and there will be a delay before this entry is activated. During the delay a user can abort auto-start with Esc, Tab, F4 keys. If any of these keys are pressed during the

countdown, the auto-start will be skipped and the main menu will be shown. Pressing **Space** will skip countdown and start the selected directory entry. During the boot sequence with auto-start the following joystick button actions are possible:

A	skip countdown and start selected entry
B	cancel auto-start

Please hold a joystick's button for at least 1 second to cancel the auto-start and go to the main menu or to start the selected directory entry. This works the same way when the message about the incompatible Boot Menu is shown.

Starting from version 2.50 of the Boot Menu, the auto-start delay is configurable via the Configuration Menu. A user can set the delay from 0 to 9 seconds. When 0 is selected, the entry configured for auto-start will be launched immediately after reboot. To get into the Main Menu it's necessary to hold **F4** key until the Main Menu appears.

VDP frequency

The **F** button only temporarily changes the VDP frequency to the desired value. Use the “Frequency at startup” setting in the [Configuration screen](#) to control what frequency you would like the computer to boot with and what frequency should be used to start ROM images. The frequency setting is saved into the configuration EEPROM and is restored even after the computer is powered on again.

CPU mode

The Turbo mode can be enabled with the **T** button only on Panasonic MSX2+ computers and R800 mode can be enabled only on Panasonic Turbo-R computers with the same button. On other computers this functionality does not work. The status of the Turbo or R800 mode is not saved into the configuration EEPROM and it is not restored at the start of the Boot Menu after the power was switched off and back on. However the Turbo/R800 mode is still set for ROMs that require a reset to start.

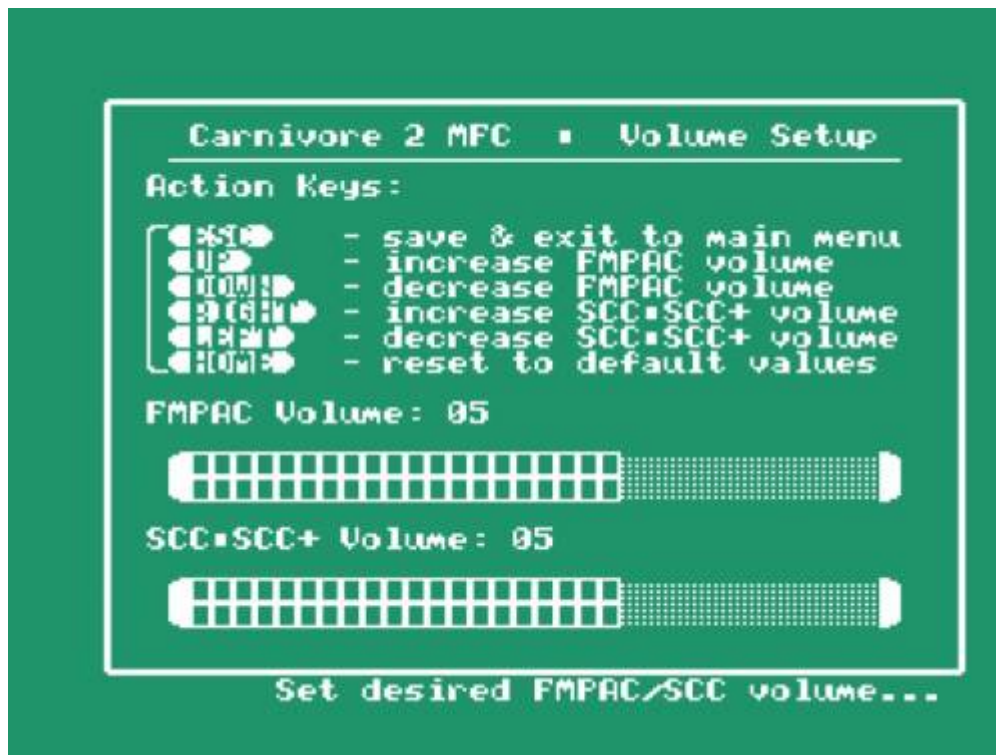
The current mode is displayed in the status screen:

Z80	Z80 mode (default)
T2+	Turbo mode for Panasonic MSX2+
R8x	R8x — R800 mode for Panasonic Turbo-R

Sound settings

The Boot Menu supports setting the volume for FMPAC and SCC sound cards. Use the **V** key to enter the volume control screen from the main menu. The following keys can be used in this screen:

Changing the volume for FMPAC and SCC sound cards:



	save & exit to main menu
	increase FMPAC volume
	decrease FMPAC volume
	increase SCC/SCC+ volume
	decrease SCC/SCC+ volume
	reset to default values

The volume's value is stored within the small EEPROM on the cartridge board. The value is saved when key is used to return to the main menu. If the small EEPROM is not present, then the volume setting is only preserved until the power-off. So setting the volume once allows to play games and listening to the music until the computer is completely switched off.

This screen also supports joystick. The joystick's movements and buttons are interpreted as follows:

	increase/decrease SCC/SCC+ volume
	increase/decrease FMPAC volume
	apply changes and exit (same as on keyboard)

It is also possible to disable the default FMPAC stereo mode using the setting in the [Configuration screen](#). This will enable the mono mode for FMPAC's sound output through the Carnivore2's audio socket.

Starting from Boot Menu v2.50, it's possible to enable the Dual-PSG mode. In this mode Carnivore2 starts to emulate PSG on alternative ports (#10 и #11). If Dual-PSG is enabled, certain games or music players may use both PSGs - one in Carnivore2 and one in MSX computer. However, it should be noted that the Boot Menu's background music will always use the default PSG ports.

PSG/PPI clicker settings

The Boot Menu also supports enabling or disabling the internal PSG and PPI's Clicker emulation as well as setting the volumes for both of them. Use the 'P' key to enter the PSG control screen from the main menu. The following keys can be used in this screen:



	save & exit to main menu
	enable/disable PSG
	enable/disable PPI Clicker
	enable/disable Dual-PSG
	increase PSG volume
	decrease PSG volume
	increase Clicker volume
	decrease Clicker volume
	reset to default values

The volume's value is stored within the small EEPROM on the cartridge board. The value is saved when key is used to return to the main menu. If the small EEPROM is not present, then the volume setting is only preserved until the power-off. So setting the volume once allows to play games and listening to the music until the computer is completely switched off.

This screen also supports joystick. The joystick's stick movements and buttons are interpreted as follows:

	increase/decrease PPI Clicker volume
	increase/decrease PSG volume
	enable or disable PSG (same as on keyboard)
	apply changes and exit (same as on keyboard)

Configuration screen

Starting from version 2.10 the Boot Menu can be customized and the custom settings will be stored in the configuration EEPROM.



To customize the configuration please use the **C** key from the main menu. Beside the cursor keys, the following keys can be used in configuration screen:

Esc	save & exit to main menu
Space	change selected value
Home	reset to default values

This screen also supports joystick. The joystick's stick movements and buttons are interpreted as follows:

A	toggle setting (same as Space on keyboard)
B	apply changes and exit (same as Esc on keyboard)

A user can customize various configuration settings including directory sorting, fade in/out effects, keyboard/joystick delay as well as the colors (font and background) for the main menu, help screen, volume control screen and PSG setup screen. In addition, a user can enable or disable the dual-reboot, disable stereo output for FMPAC and select what [frequency](#) to use at startup (50Hz, 60Hz or default VDP's frequency).

Pressing **Home** at any time will restore all customized values to default settings. Holding **F3** key at the Boot Menu's startup allows to use the default settings for the UI — all custom settings will be ignored for the current session.

Please note that editing of the palette on MSX computers with v991x or v992x video processors will be disabled. Also if the directory sorting is enabled or disabled, the current auto-start entry as well as the master/slave slot assignments are cleared to prevent a mix-up.

The directory sorting is a complex operation, so if there are many entries in the Boot Menu's directory, then it may take a few seconds to completely sort all of them. The sorting only happens at the Boot Menu's startup and when the sorting gets enabled in the configuration screen. If the delay is too uncomfortable for you, please disable the directory sorting option.

Starting from version 2.50 of the Boot Menu, 3 new options have been added: user-defined port for Carnivore2's identification and controlling, user-defined auto-start delay and the control to allow Carnivore2 to work in slot 3.

Before this version it was only possible to identify Carnivore2 in a system only by its ID marker located in the subslot 0 of the slot where the cartridge was plugged into. That was not always possible, especially when Carnivore2 was working as FMPAC cartridge, RAM expansion, IDE controller or when in SCC+ mode. Now it's possible to identify and control the cartridge via a user-defined I/O port. The following port numbers are available: #F0, #F1 и #F2. So, it's possible to find and separately control 3 different Carnivore2 cartridges in a system. More information on the I/O port can be found in the "Technical description of Carnivore2" document.

By default the port is selected automatically based on the slot number that the cartridge is located in. In this case, the port's setting has the “-” value and the currently active port number is displayed to the left from the setting. The port setting can be also changed to a fixed one - F0, F1 or F2. In this case the cartridge will always work the user-selected port number.



If there's more than one Carnivore2 cartridge in a system, each of them must be configured to use different I/O ports or the port's auto-selection has to be enabled! Otherwise there may be conflicts and a system may become unstable.

If a user allows Carnivore2 to work in slot 3, there will be no more halting a system when the cartridge is inserted into that slot. The fact is, that some MSXs have different slot assignment and slot 3 may be the second free slot available for a user. Now it will be possible to use Carnivore2 in that slot. However, it should be noted that MSXs like YIS-805, YIS-503, CX-7 and similar ones have slot 3 expanded and used by different internal devices by default. So, when Carnivore2 is used in such a slot, it will no longer work as multi-functional device. Only one of the internal devices will be able to function: FlashROM, FMPAC, RAM or IDE controller.

It's possible to set the auto-start delay in seconds. The range is from 0 to 9. If zero delay is set, then the auto-start screen will not appear at all and the menu entry assigned to be auto-started will be launched immediately. If the setting is above zero, then the auto-start screen will appear and counting from the user-defined value to zero will begin.

How to run 2 ROMs at the same time

Starting from Boot Menu's version 2.30 and the Altera's firmware version 2.30 it is possible to run 2 ROMs at the same time. This functionality is called “Dual-Slot”. The dual-slot setup screen can be called with the **O** key or by pressing **Enter** in the main menu.



The new setup screen allows to run dual-slot configuration with flexible options selection for the master slot (you can choose what Carnivore2 built-in devices to enable). You can select 2 ROMs to run simultaneously. The only restriction for the slave slot is that it can run games with Konami4 and Konami5 mappers as well as small games up to 32kb without mapper. The slave slot becomes available if there's one unused physical slot found in a computer and this slot is not occupied by another device. If there are no available slots in a system, then you can use one of the master slot's subslots to emulate the slave slot (starting from Boot Menu v2.40).

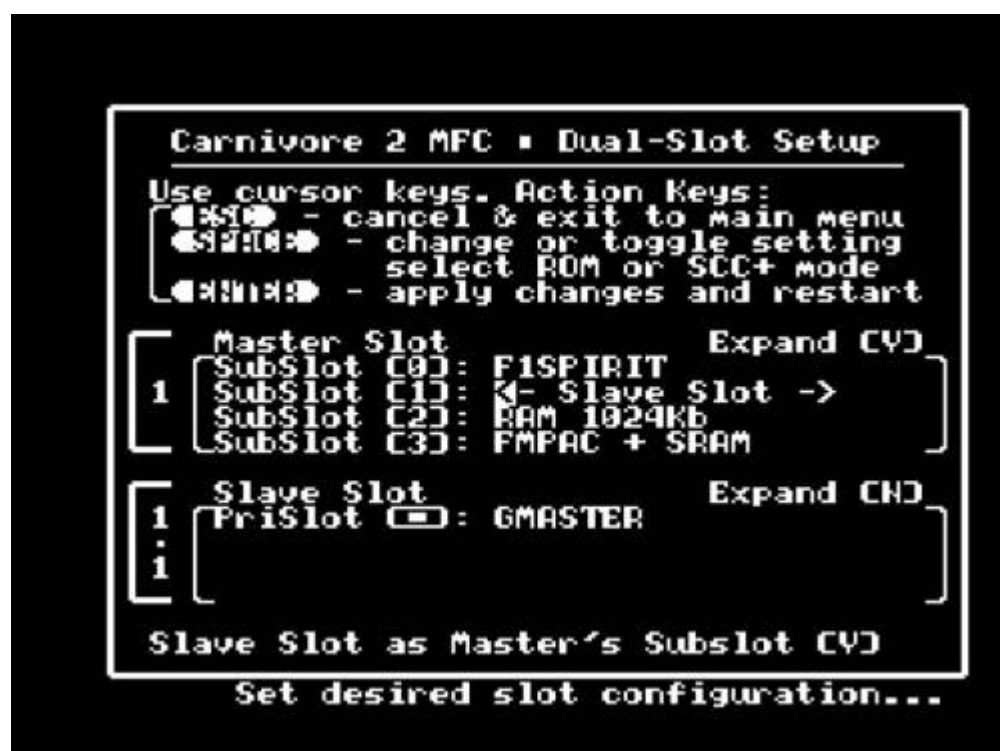
The Boot Menu identifies whether there's a suitable free slot in the MSX and then shows this slot as “slave” in the settings.

The working slots will have their numbers shown on the left side. If only one slot was identified as usable, running 2 ROMs at the same time will be possible only after enabling the “Slave Slot as Master's Subslot” option. This option is enabled automatically if no free slots are found in a system.



In order for the slave slot to work as a subslot of the master slot, the master slot must be expanded. If using the subslot of a master slot is enabled in the settings, the master slot automatically becomes expanded. If the master slot's expansion gets disabled by a user, then the “Slave Slot as Master's Subslot” option will be disabled too.

After enabling the “Slave Slot as Master's Subslot” option, you need to select what subslot of the main slot should be used to emulate the slave slot. This can be done by putting the cursor onto one of the Carnivore2's devices — RAM, IDE or FMPAC and pressing Space. The name of the device will change to “Slave Slot” and the number of the slave slot will change to N1.N2 where N1 is the number of the master slot and N2 is the number of the emulated subslot.



Be aware, that some cartridge, for example SCC, MegaRAM and other ones that do not modify the slot's area in any way will

not be detected by the Boot Menu, so the slave slot will be available even if it should not be. If Carnivore2 configures the slave slot as the same slot that is occupied by another device, this may cause conflicts and potentially damage your MSX. So please make sure that you do have enough free slots before enabling the dual-slot configuration. If you are not sure that you have a free physical slot to be used as the emulated slot, it is advised to enable the “Slave Slot as Master's Subslot” option and use one of the master slot's subslots to emulate the slave slot.

Beside the cursor keys, the following keys are usable in the Dual-Slot setup screen:

	exit to the main menu
	change or toggle setting, select ROM or SCC+ mode, enable or disable device, assign subslot for slave emulation
	apply changes and restart

The ROMs for the dual-slot configuration can be selected from the main menu. The entry for the master slot can be selected by pressing , the entry for the slave slot can be selected by pressing . The  key clears the selection as well as the Auto-Start entry. The selection will be visible in the “Dual-Slot” area above the list of ROMs, to the right from the “Auto-Start”.

Also the ROMs can be selected with the  key in the Dual-Slot screen. Pressing space bar will allow to cycle through the compatible ROMs for each slot. After the full cycle there will be an “Empty” selection to keep the slot vacant.

The “Expand” option for each slot allows to expand this slot individually. The expanded master slot allows to use all built-in Carnivore2 devices as well as to emulate the slave slot as one of the master's subslots. The slave slot can be expanded as well, but **we recommend to keep the slave slot always non-expanded to avoid conflicts with other hardware.**

In addition, the master slot can also use the “Konami SCC+” configuration. So you can run SCC+ games in the slave slot and enjoy SCC+ and a game on a single cartridge. It is recommended to use SCC+ mode only with certain games that support it. It should be kept in mind that “Konami SCC+” selection makes it impossible to use the built-in Carnivore2 devices — RAM, IDE and FMPAC as well as to use the master slot's subslot to emulate the slave slot. So to run the SCC+ configuration with a ROM image, the ROM image must be selected at the slave slot and the slave slot must be standalone.

This screen also supports joystick. The joystick's stick movements and buttons are interpreted as follows:

	same as cursor keys
	apply selected configuration and restart
	change or toggle setting, select ROM or SCC+ mode, enable or disable device, assign subslot for slave emulation (same as  on keyboard)
	exit to the main menu (same as  on keyboard)

Help screens

The two help screens provide a complete list of keys and their assignments as well as information on the joystick's stick and key assignments. By pressing any key you go to the next screen and on the last help screen pressing any key will return you to the [main menu](#).

Carnivore 2 MFC ■ Help Page 1 of 2

Menu Navigation and Action Keys:

- [F1] - boot MSX using the default configuration: all enabled
- [Left], [Right] - previous/next page
- [Up], [Down] - select ROM/CFG entry
- [Enter] - start selected entry
- [F2] - start an entry directly
- [F3] - reset MSX and start an entry
- [F4], [F5] - Dual-Slot setup page
- [F6] - select entry for master slot
- [F7] - select entry for slave slot
- [F8] - select entry for Auto-Start
- [F9] - clear Auto-Start & Dual-Slot
- [F10] - select 50Hz or 60Hz frequency
- [F11] - toggle Turbo or R800 mode
- [F12] - customize configuration
- [F13] - jump to the last used entry
- [F14] - toggle background music
- [F15] - toggle help scroller

Press any key to continue...

Carnivore 2 MFC ■ Help Page 2 of 2

Boot Option Keys:

- [F1] - use default UI settings
- [F2] - cancel Auto-Start
- [F3] - skip Boot Menu

Joystick Directional Controls:

- [Left], [Right] - same as cursor keys
 - [Up], [Down] - same as cursor keys
- NOTE: Other directions are ignored

Joystick Button Controls:

- [Button 1] - toggle (same as SPACE)
- [Button 2] - cancel (same as ESC)

Joystick Action on Auto-Start:

- [Button 1] - skip countdown
 - [Button 2] - cancel Auto-Start
- NOTE: Hold button for over 1 second

Press any key to return...

Also the help scroller can be enabled in the main menu by pressing the **S** button.

Notes for SCC+ mode

The Carnivore2 cartridge supports both SCC and SCC+ modes. Certain games started from the cartridge's IDE device may not like the SCC+ being in the expanded slot, so there will be no sound. In this case such games can be started from a different IDE device and the Carnivore2 cartridge can be configured as the SCC+ sound cartridge. To do this a new configuration entry must be created. It's necessary to start the [C2MAN](#) or [C2MAN40](#) utility, enter the directory editing mode and do the following:

1. Edit the first configuration entry "DefConfig: RAM+IDE+FMPAC+SCC"
2. Rename it to "Config: SCC+ Cartridge"
3. Select "Save/load register preset" and then choose "Load register preset file"

4. Load the provided SCCPLUS.RCP file by typing SCCPLUS and pressing  key
5. Save the configuration entry and exit the utility

Put the Carnivore2 cartridge that you want to use as the SCC+ device into the first MSX slot and the device to load games from into the second slot. Start your MSX and when the Carnivore2's cartridge menu appears, select the newly created "Config: SCC+ Cartridge" entry.

The computer will reboot and start loading a game or an operating system from the device in the second MSX slot. If the device in the second slot is configured to load a game that uses SCC+ (for example Snatcher), it will use the Carnivore2 cartridge working as SCC+ device for the output.

This functionality has been verified to be working with "Snatcher" and "Konami Game Collection" volumes 1-4 and also with the special game volume.

Alternatively, the SCC+ mode could be enabled in the [Dual-Slot](#) configuration screen. This allows to enable the SCC+ mode and run a ROM file in the slave slot (dual-slot configuration) at the same time. It is also possible to set the SCC+ configuration for the master slot. A computer will need to boot from another device because Carnivore2's IDE device will not be enabled. This way you can run disk games with SCC+ mode of Carnivore2. Please see the "[How to run 2 ROMs at the same time](#)" section for more info.

Utilities

 [Utilities in repository on GitHub](#)

C2MAN



If you find it difficult to add ROM images into Carnivore2 using the C2MAN and C2MAN40 command line utilities, please switch to using SofaRun. This software has native support for Carnivore2 and is very user-friendly. The software can be downloaded from here:

<https://www.louthrax.net/mgr/sofarun.html>



If you have an Arabic or Korean MSX computer, you should use the special version of the C2MAN or C2MAN40 utility that can be downloaded from here:

- [C2MAN](#)
- [C2MAN40](#)



If utilities C2MAN and C2MAN40 don't run correctly or MSX-DOS shows an insufficient memory error, try **C2MINI** or **C2MINI40** utilities. These are smaller versions of the original utilities, but without the possibility to edit directory entries. Deletion of entries, however, is still possible.

The C2MAN utility allows to initialize the cartridge, add ROMs into the FlashROM, create custom configuration entries, edit the cartridge's directory. The [Service Menu](#) allows to see the [FlashROM block usage](#), erase and [optimize the directory](#), upload the Boot Menu as well as IDE and FMPAC BIOSes into the FlashROM; it also allows to completely erase the FlashROM chip.

The C2MAN utility works only on MSX2 and later computers, it sets the 80 character mode by default. On MSX1 computers this utility shows an incompatibility note and exits.

For MSX1 computers the C2MAN40 utility must be used. This utility, however, will also work on MSX2 and later computers in 80 character mode, but all messages will be truncated for the 40 character mode.

Both utilities will automatically reboot a computer after uploading a ROM into the FlashROM chip if the /a and /r command line options are used.

The utility supports the following command line options:

```
C2MAN [filename.rom] [/h] [/v] [/a] [/r] [/su]
```

- /h — help screen
- /v — verbose mode (show detailed information)
- /a — automatically detect and write ROM image (no user interaction needed)
- /r — automatically restart MSX after flashing ROM image
- /su — enable Super User mode (allows editing all registers and overriding IDE BIOS write lock when BIOS shadowing is off)

The utility is normally able to find the inserted cartridge by itself. If the utility can't find the cartridge, you will need to input the slot number manually and press Enter ↵. The slot number is “10” for first slot, “20” for second slot, and so on.

Main menu

List of menu options:

```
Main Menu
-----
1 - Write ROM image into FlashROM
2 - Create new configuration entry
3 - Browse/edit cartridge's directory
4 - Restart the computer
9 - Open cartridge's Service Menu
0 - Exit to MSX-DOS [ESC]
```

The main menu allows to:

<u>1</u>	write new ROM image into FlashROM
<u>2</u>	create new configuration entry
<u>3</u>	browse/edit cartridge's directory
<u>4</u>	restart a computer
<u>9</u>	open cartridge's service menu
<u>Esc</u>	exit to MSX-DOS

The menu options should be selected with the corresponding numeric buttons.

Adding a ROM file into the FlashROM

To add a new ROM file into the FlashROM chip, select the “Write new ROM image into FlashROM” option. Follow the on-screen instructions until the ROM is successfully written into the chip and the main menu re-appears. The large ROMs' mappers should be normally detected automatically by the utility, but on some ROMs autodetecting may fail. In this case the utility will ask you to choose the mapper. The ROM will not start with incorrect mapper settings, so if your setting didn't work, try to change the mapper type.

The FlashROM chip contains 128 blocks by 64kb (8mb in total). The first 4 blocks are occupied by the Boot Menu, directory, IDE BIOS and FMPAC BIOS. Other blocks are available for a user to add the ROMs. The ROMs that are smaller than 64kb are grouped into one block. For example two 32kb ROMs will be written into the same 64kb block, eight 8kb ROMs will be grouped into the same 64kb block and finally four 16kb ROMs will be grouped written into the same 64kb block. All this is done automatically.

You can add a ROM into the chip without user interaction. The following command line should be used:

```
C2MAN file.rom /a
```

The utility will try to automatically detect the ROM's mapper, check whether any free space is available and then it will write the selected ROM into the FlashROM chip. If you add the /v option, the utility will show additional information about the chip and the ROM that is being added as well as the map of the free chip's blocks.

The map of FlashROM chip blocks can be viewed from the “[Service Menu](#)”. Just select the “[Show FlashROM chip's block usage](#)” option.

Adding a custom configuration entry

To add a new configuration entry select the “Create new configuration entry” option. You will be asked to enter the name of the entry and then you will need to answer 5 questions:

```
Enable extended slot? (y/n)
Enable RAM and Mapper? (y/n)
Enable FMPAC? (y/n)
Enable IDE controller? (y/n)
Enable SCC and MultiMapper? (y/n)
```

The utility will ask:

- whether the slot should be expanded or not (if you want to enable more than one internal device, the slot must be expanded),
- whether to enable one or more of the 4 built-in devices:
 - RAM + mapper,
 - FMPAC,
 - IDE controller
 - MultiMapper + SCC.

You can select any combination you want. The cartridge can work as pure SCC or FMPAC sound cartridge, as a 1 MB RAM expander or as a disk drive. Or as a combination of those devices.

The configuration entries will have the “C” symbol close to their names. Once the configuration entry is selected, the MSX will restart to take the new configuration into effect.

The configuration entries don't occupy any space in the FlashROM chip, so they can be created as long as there's free space in the cartridge's directory.

Editing or deleting directory entries

To edit the cartridge's directory select the “Browse/edit cartridge's directory” option. This will open the screen with the list of directory entries, 10 per page. The key assignment is similar to the Boot Menu with the exception that you can't start the entry.

An entry can be edited or deleted. Follow the on-screen instructions for editing a directory entry. Please keep in mind that the very first entry called “DefConfig: RAM+IDE+FMPAC+SCC” can't be deleted.

In the directory editor you can change almost all fields of an entry, select a different mapper, enable or disable the internal devices or expanded slot (some games don't like being in the expanded slot). The editor has the context based help that is displayed at the bottom of the screen.

With the Super User mode you can edit any register you want, but be advised, that you may damage the directory beyond repair and you will need to initialize it to continue using the cartridge.

When you finish editing, you need to save the entry. The utility will offer you to replace the older entry or to create a copy of the edited entry. The new entry will be located in the end of the list. The name of the entry will be the same if you didn't rename it while editing.

The number of directory entries is limited to 254. If the utility can't find an empty directory entry, it will ask you whether the

directory should be [optimized](#). If you select “Yes”, then there's a big chance that unused directory entries will be found and deleted and you will have the possibility to add new ones.

Loading and saving RCP files

The RCP stands for “Register Configuration Preset”. It is a small data file with the settings for certain non-standard games or configurations. The RBSC provides many RCP files for the most popular games and software in the repository, in the Presets subfolder.

When a ROM file doesn't start properly after being detected by the C2MAN utility, there may be a need to adjust its configuration. This can be done either manually — by editing the configuration registers or by loading an RCP (Register Configuration Preset) file. We are providing a few RCP files for the ROM files that are not working correctly with default configuration.

To load the RCP file manually you need to run the C2MAN utility, enter the directory editor and start editing the selected ROM entry. When editing, select the “Save/load register preset” option and then use “Load register preset file”. When asked, enter the preset's file name and it will be loaded for the entry you are editing. Just save the entry with the new settings and your ROM will start working correctly.

When you are making your own configuration settings for a selected ROM file, you can always save them into RCP file. You need to select the “Save/load register preset” option and then use “Save register preset file”. When asked, enter the name of the RCP file and it will be saved for future use.

The latest versions of C2MAN, C2MAN40 and C2RAMLDR utilities try to automatically find the matching RCP file when a ROM is being loaded. For example if a user writes the TEST.ROM file into the cartridge, the utilities will try to locate the TEST.RCP file and ask a user whether he/she wants to load and use the data from that RCP file. When a ROM file is loaded with the /a command line option, the data from the matching RCP file is automatically applied.

Service menu

To enter the cartridge service menu, press [9](#) in [Main menu](#).

List of menu options:

```
Service Menu
-----
1 - Show FlashROM's block usage
2 - Optimize directory entries
3 - Init/Erase all directory entries
4 - Write Boot Menu (BOOTCMFC.BIN)
5 - Write IDE ROM BIOS (BIDECMFC.BIN)
6 - Write FMPAC ROM BIOS (fmpcmfc.bin)
7 - Fully erase FlashROM chip
0 - Return to main menu [ESC]
```

The following options are available here:

1	show FlashROM's block usage
2	optimize directory entries
3	init/erase all directory entries
4	write Boot Menu
5	write Nextor IDE BIOS
6	write FMPAC BIOS
7	fully erase FlashROM chip
0 , Esc	return to main menu

Show FlashROM's block usage

Map of FlashROM chip's 64kb blocks (FF = reserved, 00 = empty):

	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F
00		FF	FF	FF	FF	FF	00	00	00	00	00	00	00	00	00	00
10		00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
20		00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
30		00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
40		00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
50		00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
60		00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
70		00	00	00	00	00	00	00	00	00	00	00	00	00	00	00

Press any key to continue

Optimize directory entries

Optimization of directory entries updates the default configuration entry "DefConfig: RAM+IDE+FMPAC+SCC" so that the cartridge could be detected by the utilities. This also removes the gaps between the entries. This operation should be performed, for example, after deleting a few directory entries.

Init/Erase all directory entries

The service menu option "Init/Erase all directory entries" allows you to delete all directory entries. In this case, the correct DefConfig record is automatically created, which is necessary for the correct detection of the cartridge by utilities.

Write Boot Menu

This service menu option updates the Boot Menu program. Without this program, the cartridge will work as the IDE controller and RAM module, FM Basic will also be available. With the missing Boot Menu, it will be impossible to start ROMs as well as configuration entries. The Boot Menu is one of the most important programs for Carnivore2. To write the Boot Menu, you need the B00TCMFC.BIN file

The latest version is available [here](#).



If you have an Arabic or Korean MSX computer, you should use the special version of the Boot Menu that can be downloaded from [here](#).

Write Nextor IDE BIOS

This service menu option writes the ROM of the disk controller into the cartridge. Disk drive functionality will not be available without this component. Updating Nextor IDE BIOS requires the BIDECMFC.BIN file.

The latest version is available [here](#).

Write FMPAC BIOS

This option of the service menu writes the FMPAC BIOS into the cartridge so that the FMPAC sound card is properly detected by different programs and that the FM Basic could become available. To update the FMPAC BIOS you need the FMPCCMFC.BIN file

The latest version is available [here](#).

Fully erase FlashROM chip

This option allows the contents of the FlashROM chip to be completely erased! After this operation, the cartridge will be unusable until the Boot Menu and at least the IDE BIOS are written into it and until the directory is initialized. This will be possible only after booting into MSX-DOS from another cartridge, for example, from the Sunrise IDE controller (Maxiol or similar) or from a floppy drive or Gotek emulator. If you cannot start MSX-DOS from another device, do not erase the FlashROM!

C2RAMLDR

C2RAMLDR.COM — the utility to load ROMs into the cartridge's RAM.

It is possible to use the cartridge as a MegaRAM — for loading ROM images into the cartridge's own RAM and starting them after reboot. The C2RAMLDR.COM utility allows to copy ROM images up to 1mb into the cartridge's RAM and it also creates a directory entry for the copied ROM with the "RAM:" prefix before the name.

This utility is similar to [C2MAN.COM](#) utility — it has a menu that allows user to select copying the ROM image into RAM with or without protection. If the ROM is copied without protection, it will be able to write into its own address space. Some games that have copy-protection will corrupt their data and won't work. So it's always recommended to apply protection for the copied ROM image in RAM. The utility can be also used from the command line to automatically load the ROM image into RAM without any user interaction.

The utility has a feature to restart a computer after loading a ROM image into the cartridge's RAM. This can be either done from the utility's main menu or by specifying the /r command line option together with the /a option.

Please note that the ROM's image exists in the cartridge's RAM only until the next power-off unless there's a battery installed onto the cartridge's board to always preserve the RAM's data. Don't power-off your MSX if you want to keep the ROM in the cartridge's RAM. The "RAM:" entries are also selectable in the Dual-Slot setup screen. They will work only while the power is on.

The old directory entries with "RAM:" prefix, created by the [C2RAMLDR.COM](#) utility can be deleted by the same utility during the directory's optimization, they can be also removed by [C2MAN.COM](#) or [C2MAN40.COM](#) utility in the editor mode and by optimizing the directory. After power-off these directory entries become useless anyway.

C2SRAM

C2SRAM.COM — the utility for managing the FMPAC's SRAM data.

The FMPAC's 8kb SRAM is emulated by the cartridge at the Shadow RAM's address 0FE000h. This area is not affected by the 1mb of primary RAM in any way. This area is used by certain games to save the data. If the Carnivore2 cartridge doesn't have a backup RAM battery installed, then the data that was saved into that area will be lost when an MSX computer is switched off. As this data survives the reset, it's possible to save it to a file and load it back into RAM when needed. The utility that allows to save/load this data is called C2SRAM.COM. The files with the save data will have .SRM extension by default and these files will be found by the utility when a user selects files manually. However the file can be saved with any name and extension. In such a case when a user wants to upload the file into the emulated SRAM area, he will have to type its name manually.

To save the data it's enough to reset MSX (no power-off!), run the "C2SRAM.COM" utility and save the data to a file. Then a computer can be switched off. In case a user wants to restore the data and then run a game, the C2SRAM.COM utility should be

used to upload the previously saved file into SRAM area. Then a computer should be reset and a game can be then started from the Boot Menu or from an emulated DSK image.

C2BACKUP

C2BACKUP.COM — the utility to backup and restore the contents of the FlashROM chip.

The C2BACKUP.COM utility allows to dump the contents of the entire FlashROM chip into a file. The size of the file is 8 MB, the time required for the operation is about 10 minutes. The utility also allows to copy the contents of the FlashROM's dump back into the chip. Because of the BIOS shadowing this operation can be performed live, however the system must be restarted as soon as possible after uploading the new contents into the FlashROM chip.

The utility also allows you to copy the contents of the FlashROM image back to the chip. After filling the FlashROM image into the chip, you need to restart your computer. It is possible that after restoring data from a backup you will need to update:

- [IDE BIOS](#)
- [FMPAC BIOS](#)

List of menu options:

```
Main Menu
-----
1 - Download FlashROM's contents to a file
2 - Upload file's contents into FlashROM
3 - Restart the computer
0 - Exit to MSX-DOS
```

The following features are available here:

1	save folder contents from FlashROM to a file
2	restore directory contents from a file to FlashROM
3	restart the computer
0 , Esc	exit to MSX-DOS

The utility asks a user whether he would like to preserve the existing Boot Menu on the cartridge and in case of a positive answer it doesn't overwrite the existing Boot Menu with the one stored in the backup file. In this case the utility shows the «-» symbol instead of “>” when skipping writing of the Boot Menu.



WARNING! Interrupting the FlashROM's contents uploading may result in a bricked Carnivore2 cartridge! In this case the cartridge must be re-initialized. The description of the procedure can be found in the [“How to enable the cartridge and install BIOS ROMs”](#) section.

C2CFGBCK

C2CFGBCK.COM — the utility to backup and restore configuration settings.

The C2CFGBCK.COM utility allows to dump the contents of the configuration EEPROM chip into a file. The size of the file is 128 bytes. The utility also allows to copy the contents of the EEPROM's dump back into the chip. The system must be restarted after uploading the new data into the EEPROM chip for the configuration changes to be taken into use.

C2IDETST

C2IDETST.COM — utility is used to test IDE/FDD controller's read/write functionality. The program runs the required number of tests and displays the results of their execution.

Starting with version 1.05, the program for the program has the ability to specify the drive letter and the number of test repetitions.

Syntax:

```
C2IDETST [/?] [/N] [Drive]
```

where

- N — number of iterations, from 2 to 99
- Drive — drive letter, A-Z
- ? — show help

Examples:

C2IDETST /25 A	run 25 iterations of test for drive A
C2IDETST /?	show help

When the program is launched without parameters, 10 read/write tests are performed for the current disk.

To stop the test it's necessary to hold the `Esc` key. In the end the utility shows the total/success/failed counters. At the end of testing, the program will shows the passed/failed status of any of the disk operations fail.

C2FINDER

C2FINDER.COM — utility to find Carnivore1 and Carnivore2 cartridges in a system using two different methods. The utility shows FlashROM's information, including its technical data (if /v or “detailed” menu option is used).

The following menu options are available:

```
Main Menu
-----
1 - Find Carnivore cartridges (brief)
2 - Find Carnivore cartridges (detailed)
3 - Restart the computer
0 - Exit to MSX-DOS
```

The following options are available:

<code>1</code>	show all Carnivore cartridges in a system
<code>2</code>	show all Carnivore cartridges in a system with extended information
<code>3</code>	restart the computer
<code>0</code> , <code>Esc</code>	exit to MSX-DOS

The utility supports several command line options.

Syntax:

```
C2FINDER [/b] [/v] [/h] [/r]
```

where

- '/h' — show help screen
- '/b' — basic mode (show basic information)
- '/v' — verbose mode (show detailed information)
- '/r' — restart the computer

Examples:

C2FINDER /b	Detect Carnivore and show basic information
C2FINDER /v	Detect Carnivore and show extended information

Carnivore2 support in openMSX

The openMSX emulator supports Carnivore2 emulation from version 0_14_0-200. To configure the emulation, you will need special files that can be downloaded from the link below.

To add Carnivore2 device into openMSX please do the following: Put files Carnivore2.xml and Carnivore2.rom into the emulator's folders as specified below:

File	Target path for	
	MS Windows	Linux
Carnivore2.xml	\openMSX\share\extensions\	/usr/share/openmsx/extensions/
Carnivore2.rom	\openMSX\share\systemroms\other\	/usr/share/openmsx/systemroms/other/

Run "openMSX Catapult", select Settings, click Edit configuration and [OK](#).

The device called Carnivore2 will appear in the list of the found devices. You can then attach a disk image to Carnivore2 by specifying the location of the DSK file (your own CF card's image) in the Catapult's user interface. Click on the [Hard Disk](#) button and locate the desired image file.

If you already have Carnivore2 in your openMSX and you only want to update the FlashROM, you may copy the Carnivore2.rom file into this folder as carnivore2.flash:

- MS Windows

```
C:\Users\<user_name>\Documents\openMSX\persistent\Carnivore2\untitled1\
```

where <user_name> is your Windows user name;

- Linux

```
~/openMSX/persistent/Carnivore2/untitled1/
```

Please be advised that **all your previous data on the FlashROM will be gone!** So if you want to preserve the data, but to have the latest Boot Menu and IDE BIOS versions, you need to run openMSX, boot to MSX-DOS and use the [C2MAN](#) or [C2MAN40](#) utility to update the [Boot Menu](#) and [IDE BIOS](#) using the latest BIN files from the Carnivore's Github repository (in folders [BIOSes](#) and [BootMenu](#)).



Certain features of Boot Menu v2.30 and later versions may not work until the support for them is added into openMSX:

- FMPAC mono mode will not be enabled
- Dual-Slot screen will not allow to run ROMs in the slave slot
- The firmware version may not be shown correctly

The files for adding support for Carnivore2 into openMSX are available [here](#).

Troubleshooting

The cartridge has a pushbutton to completely disable its functionality if something goes wrong. If the cartridge stops working correctly, you may need to reinitialize it like described in the [How to enable the cartridge and install BIOS ROMs](#) section.

You need to boot to MSX-DOS from another disk device or a floppy drive in order to use the C2MAN or C2MAN40 utilities to initialize the Carnivore2 cartridge. A computer has to be started while holding the cartridge's pushbutton. When the DOS prompt appears, the button can be released.

Disclaimer

The RBSC provides all the files and information for free, without any liability (see the  [disclaimer.txt](#) file in the repository). The provided information, software or hardware must not be used for commercial purposes unless permitted by the RBSC. Producing a small amount of bare boards for personal projects and selling the rest of the batch is allowed without the permission of RBSC.

When the sources of the tools are used to create alternative projects, please always mention the original source and the copyright!

Credits

The following individuals have contributed to the success of Carnivore2 project:

- Ptero [RBSC]
- Wierzbowsky [RBSC]
- Greywolf [RBSC]
- DJS3000 [RBSC]
- SuperMax [RBSC]
- TNT23 [RBSC]
- Pyhesty [RBSC]

- Uniskie
- Pencioner
- SolidSnail
- Konamiman
- Mitsutaka Okazaki
- Kazuhiro Tsujikawa
- Max Iwamoto
- GDX
- Grauw
- Spark/SDM
- KOD/SDM
- Wbahnassi
- Carmeloco
- Nyyrikki
- 8bits4ever
- Robodrunk
- Vogul
- AlexBel

We would like to thank the creators of the MSX platform for their wonderful invention that brings joy to many people for more than 30 years.

Contact information

The terms and conditions for commercial production of the cartridge can be found here:

<http://rbsc.su/en/licensing/#Carnivore2>

The members of RBSC group Tnt23, Wierzbowsky, Pyhesty, Ptero, GreyWolf, SuperMax, VWarlock and DJS3000 can be contacted via the group's e-mail address:

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The group's website can be found here:

[**https://rbsc.su/**](https://rbsc.su/)

[**https://rbsc.su/ru**](https://rbsc.su/ru)

The RBSC's hardware repository can be found here:

[**https://github.com/rbsc**](https://github.com/rbsc)

The RBSC's 3D model repository can be found here:

[**https://www.thingiverse.com/groups/rbsc/things**](https://www.thingiverse.com/groups/rbsc/things)

<https://sysadminmosaic.ru/en/msx/carnivore2/carnivore2>

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